Teaching Circuit Design and Measurements



Course Materials

Software

- NI Multisim 11.0
- NI ELVISmx 4.1.1 driver
- NI LabVIEW 8.5 or later

Hardware

- NI ELVIS II/II+
- Components & Wire

Manual /Files

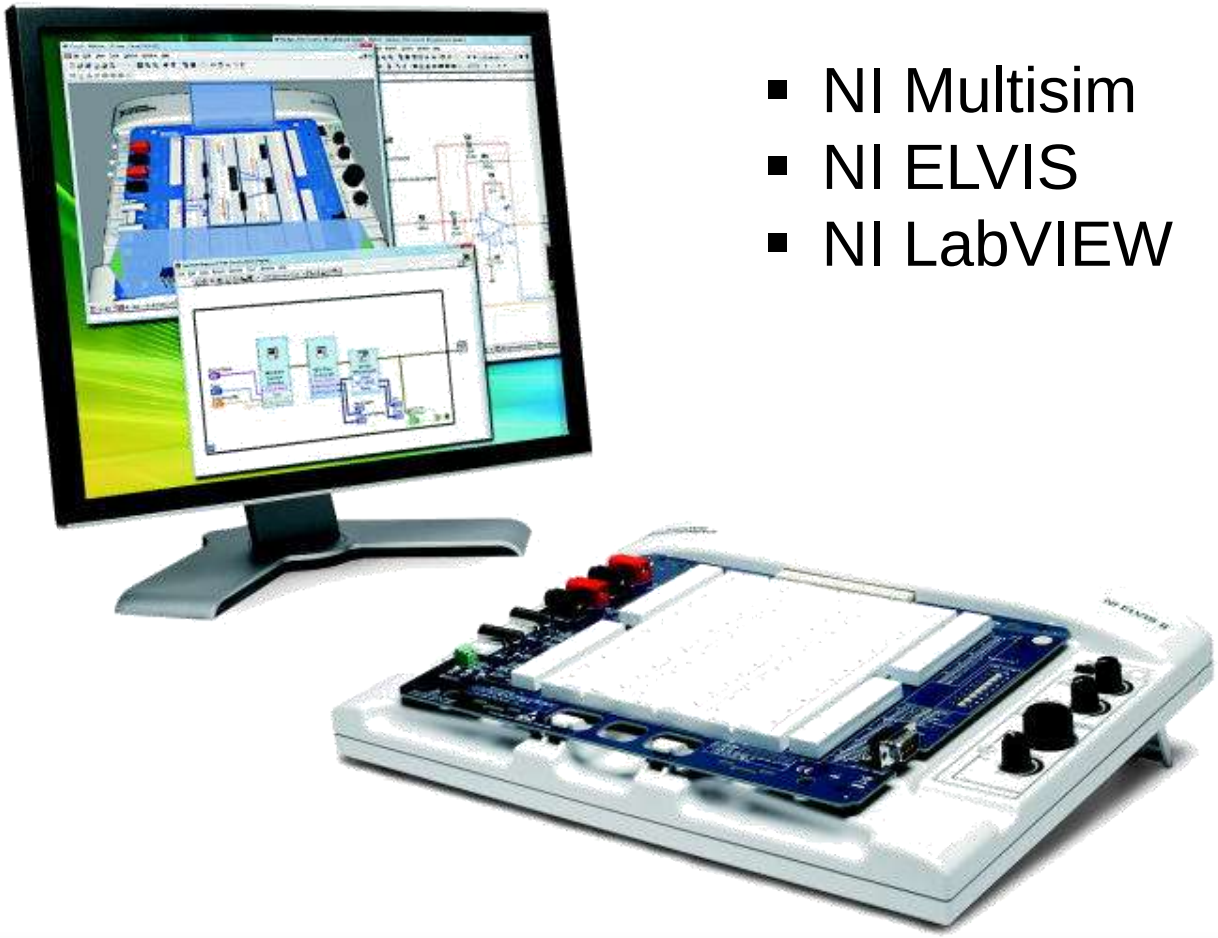
- Exercises printed material
- Sample Circuit Files



Hands-on Agenda

A. Introduction to the Components of the Circuits Teaching Platform
B. Schematic Capture
C. Simulation
D. Prototyping
   - NI ELVIS Instruments
   - Multisim and NI ELVIS Integration
E. Programming NI ELVIS with LabVIEW

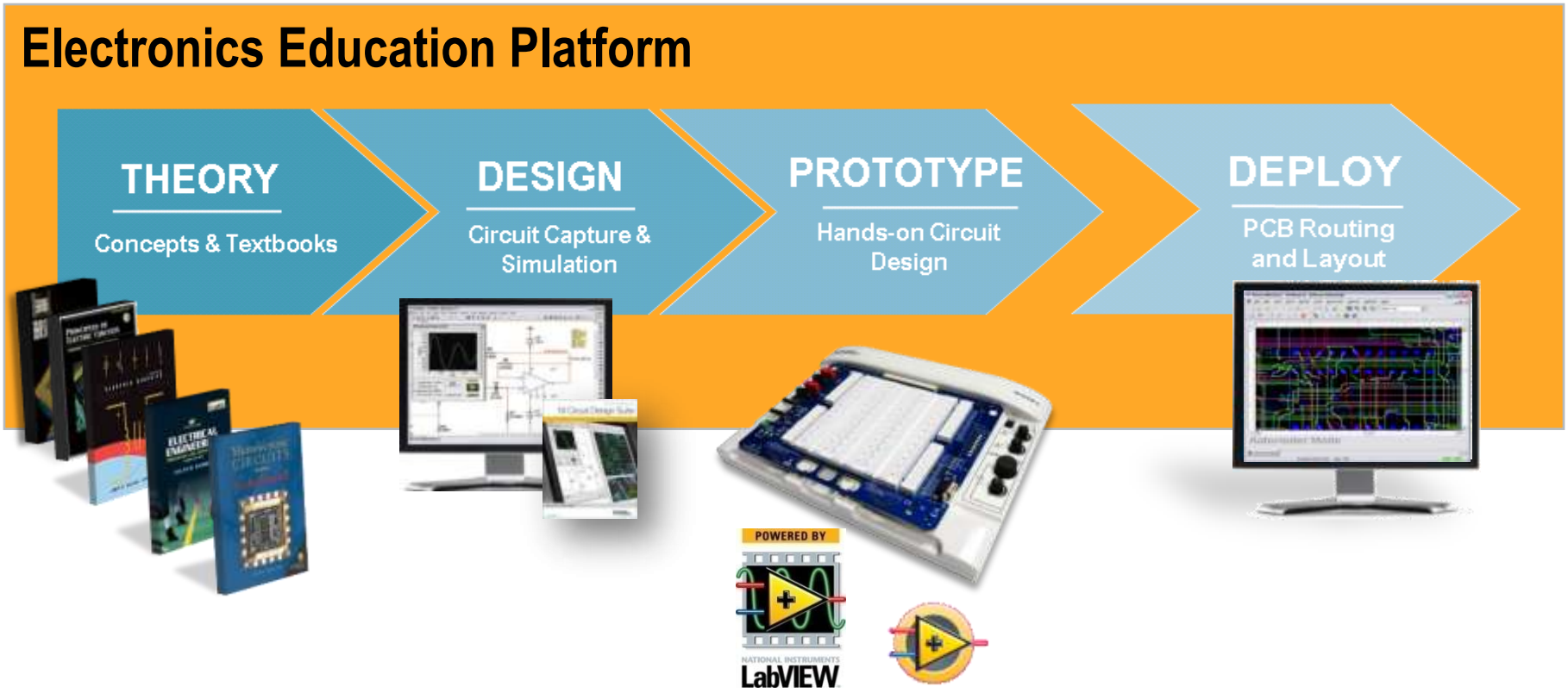
A. Introduction to the NI Electronics Education Platform



- NI Multisim
- NI ELVIS
- NI LabVIEW

A. NI Electronics Education Platform

National Instruments tools address challenges of circuit education



National Instruments | Circuit Design Suite

National Instruments Circuit Education Platform

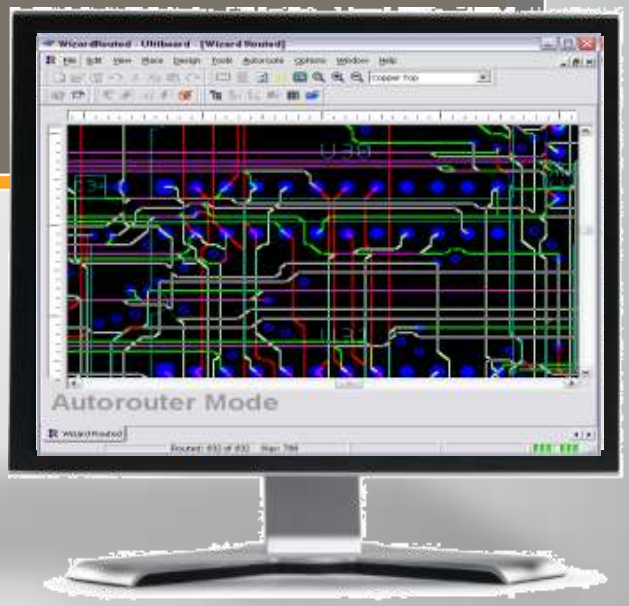
NI Multisim

A unified environment for the design and simulation



NI Ultiboard

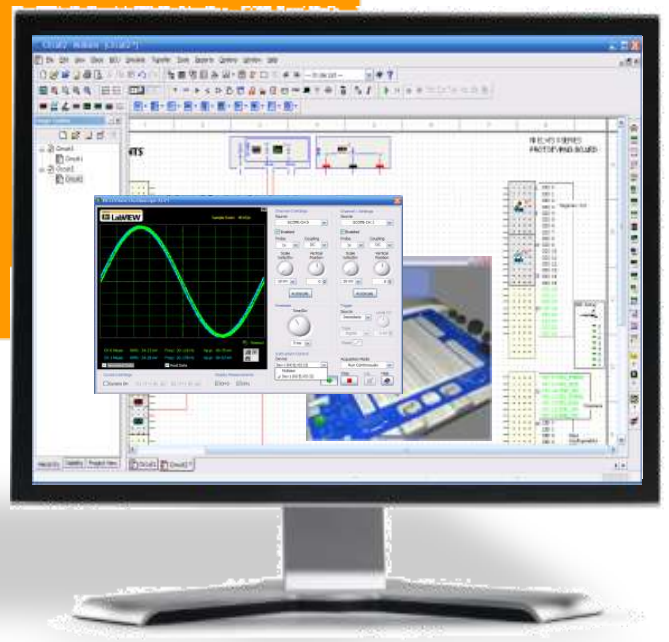
Prototype CAD design implemented



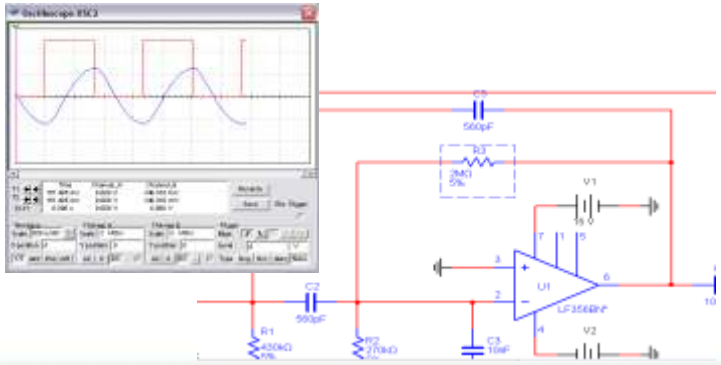
NI Multisim

NI Multisim

Teaching environment to foster student learning



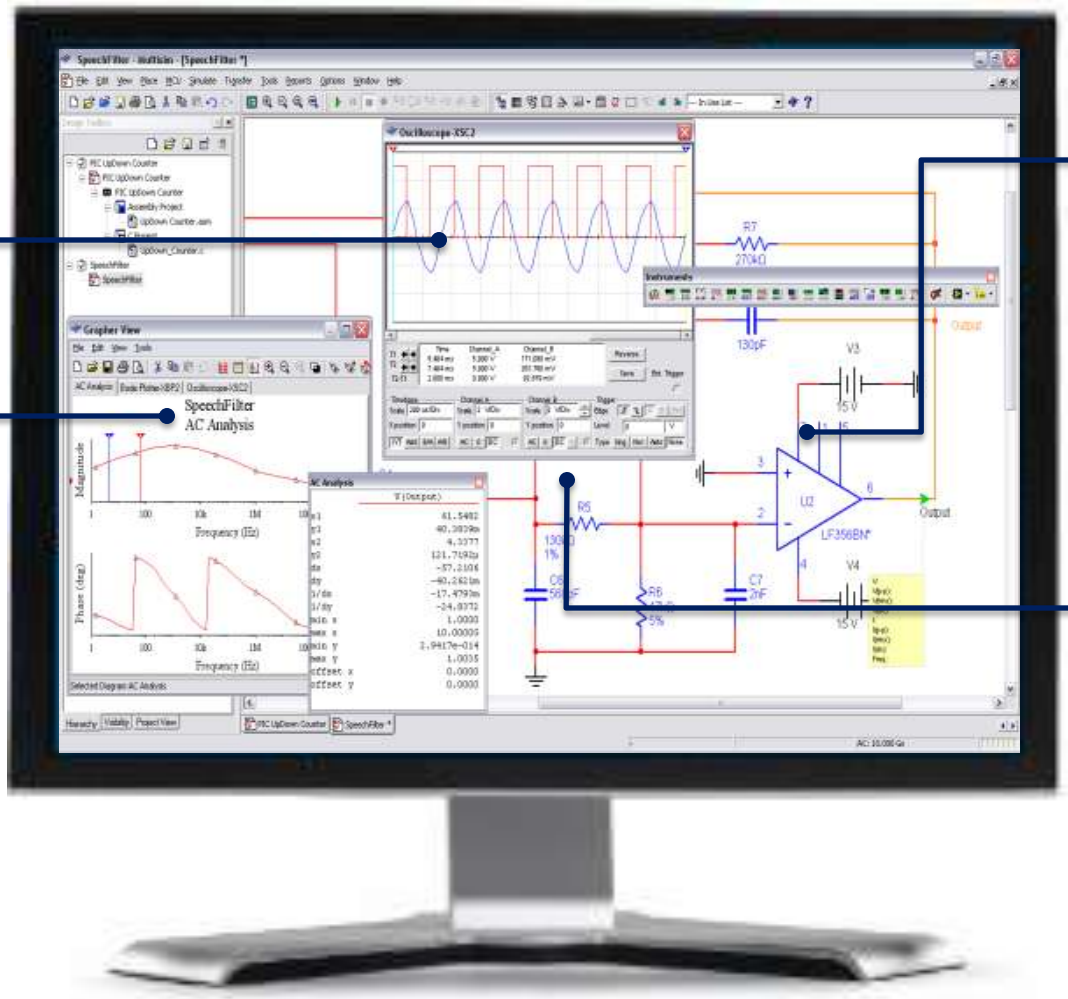
- Engage students with interactive components and change-on the-fly simulations
- Gain intuition through dynamic visualization using simulation-driven instruments
- Bridge gap between theory and real-world measurement



Core NI Multisim Educational Features

22 Virtual Instruments behaving like real-world counterparts

Advanced Analyses to investigate circuit characteristics

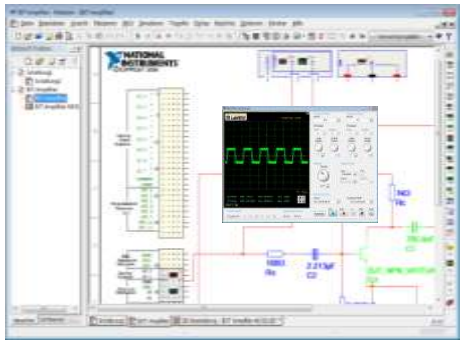


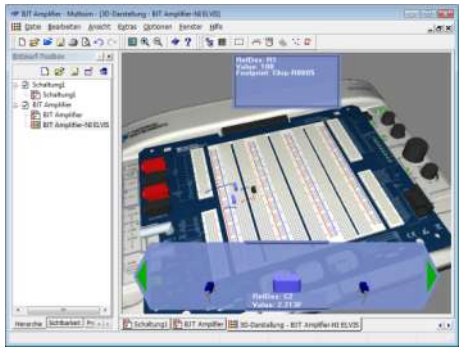
15,000+ Components to reinforce theory

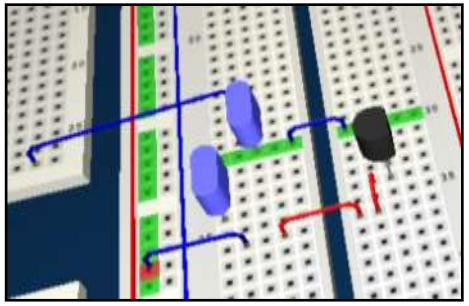
Teach troubleshooting with circuit restrictions and hidden faults

Multisim Teaching Environment

- NI Multisim
  - SPICE-based simulation
  - Analog, digital, mixed
  - Interactive parts
  - Virtual instruments
  - Circuit faults and restrictions
- Integration with NI ELVIS II/II+
  - 3D virtual breadboard
  - NI ELVIS instruments
  - Input/output of real-world signals







NI Ultiboard

NI Ultiboard

Unified environment for PCB layout and fabrication



- Easy export of Multisim designs to layout tool
- Create production-ready files
- View completed PCB in 3D
- Ideal for senior design

A. What is NI ELVIS?

Oscilloscope | DMM | Function Generator | ARB | Bode Analyzer | DSA | Power Supply

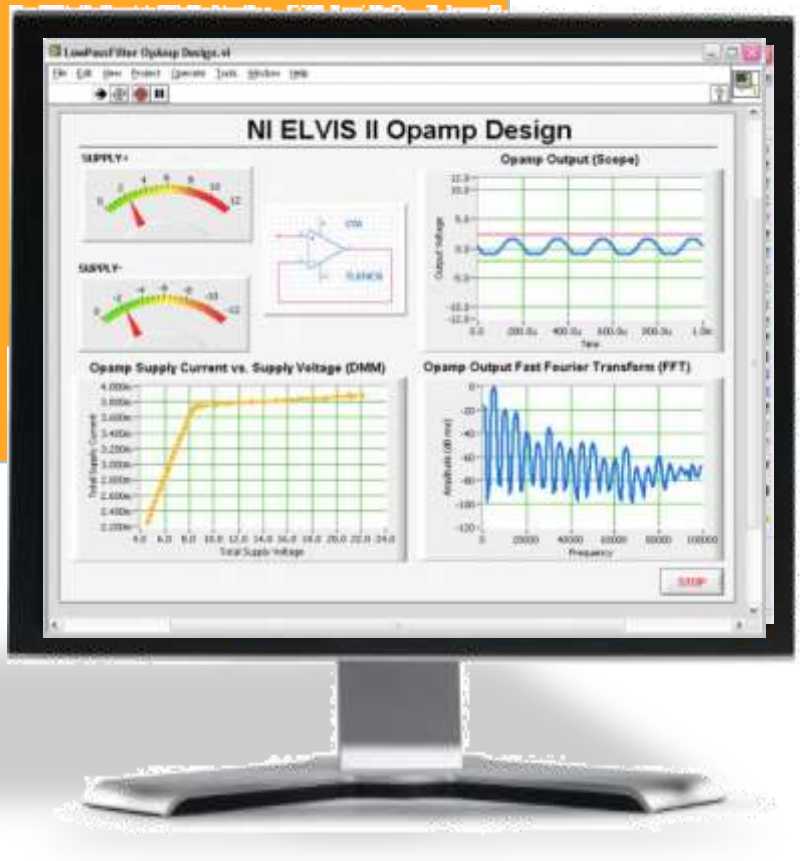


Instrumentation | **Circuits** | **Signal Processing** | **Control Design** | **Communications** | **Mechatronics**

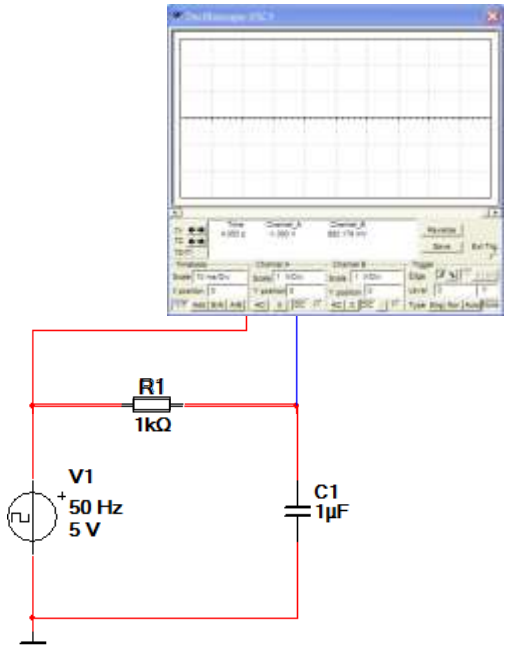
A. NI LabVIEW

NI LabVIEW

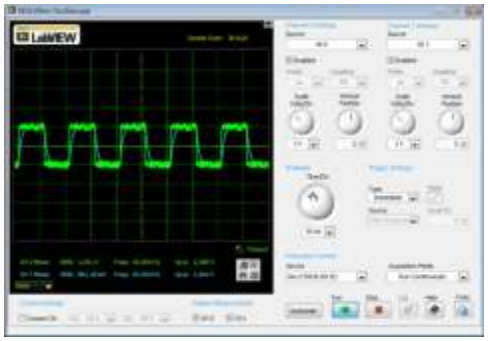
Graphical programming software for interface with measurement and control hardware, analyze data, share results, and distribute systems.



Teaching Circuits



Step 1: Capture, Simulate and improve a design in NI Multisim

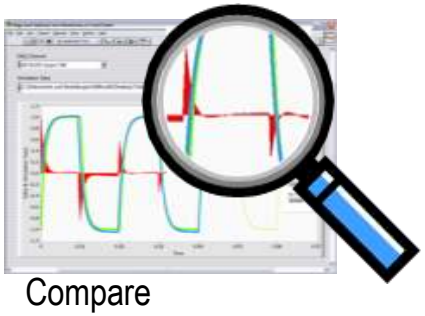




Step 2. Build circuit and measure Real-world signals with NI ELVIS



Recognize and Learn



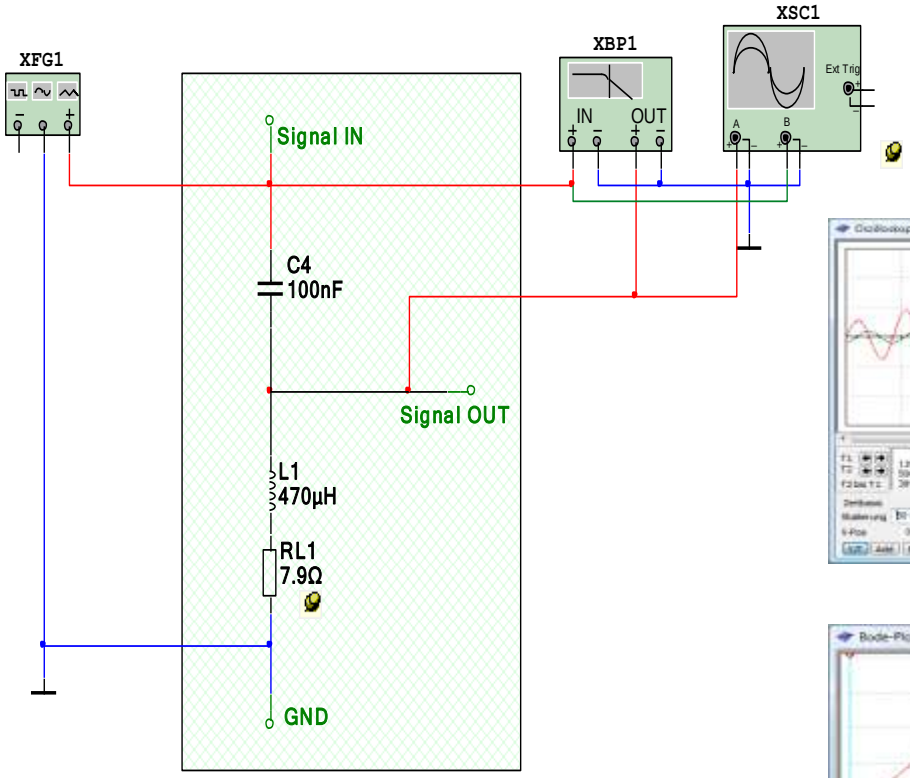
Compare

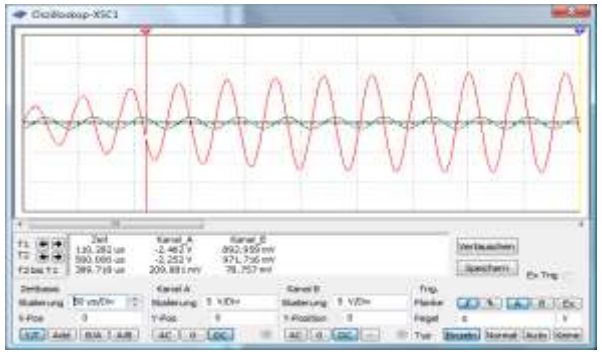
Step 3. Compare *Simulated vs. Real* Measurement Data

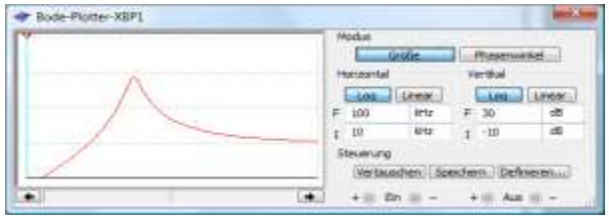
Teaching Circuits | Series RLC

Step 1: Capture, Simulate and understand a design in NI Multisim





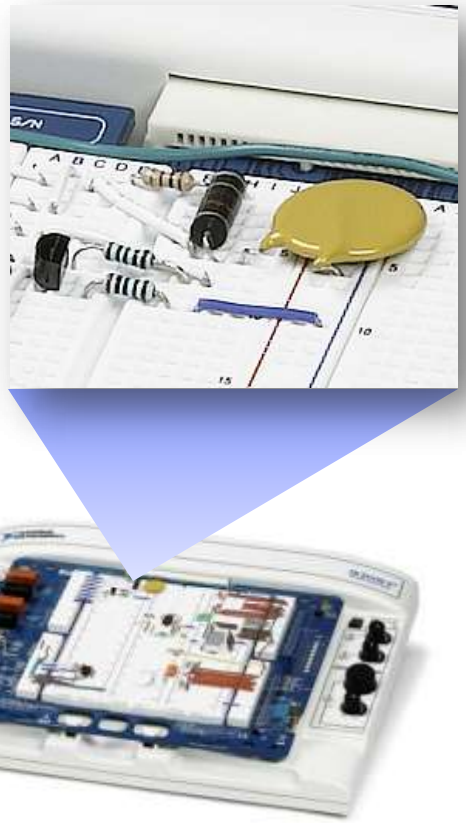


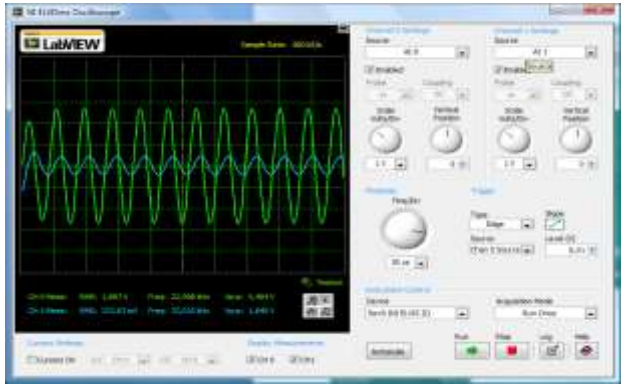


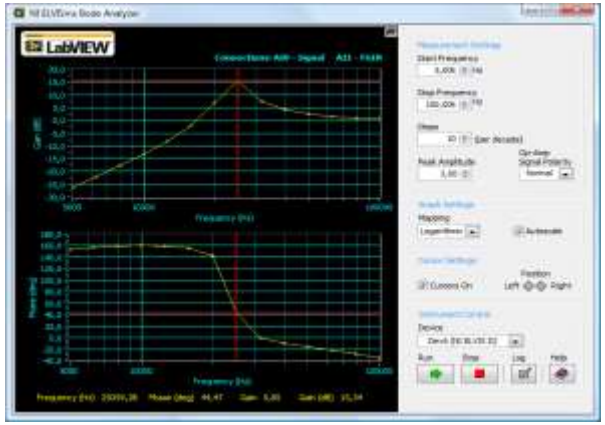
Teaching Circuits | Series RLC

Step 2: Build circuit and measure Real-world signals with NI ELVIS



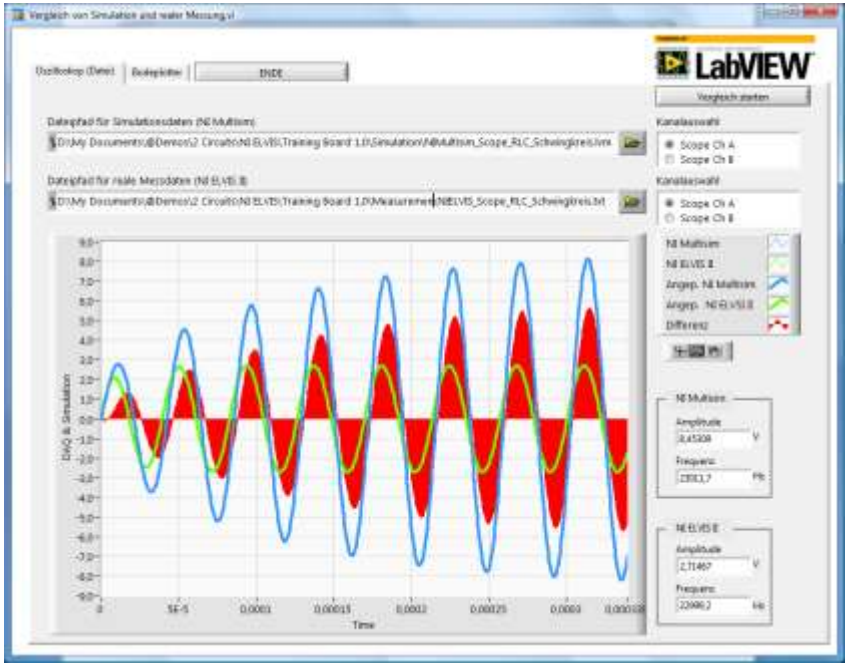




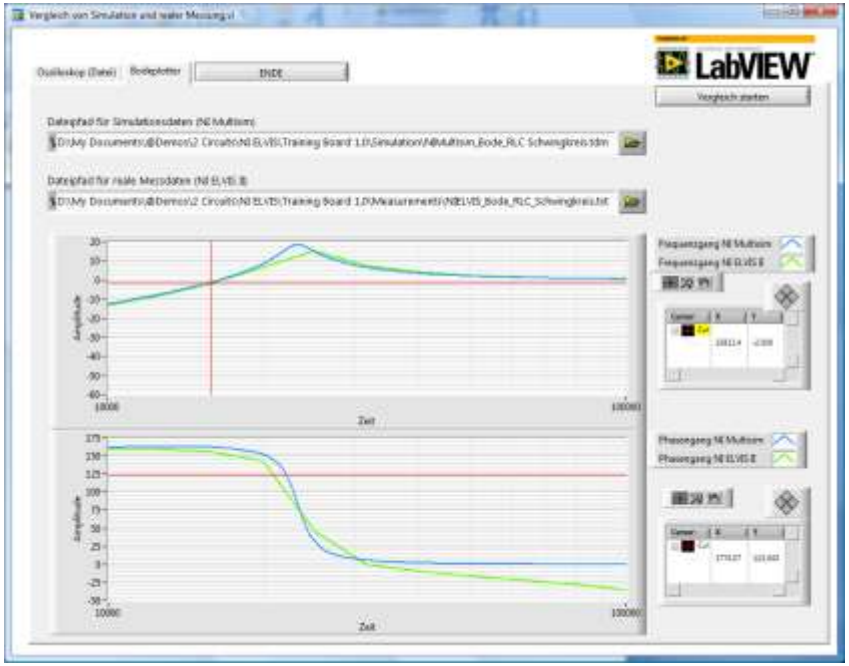


Teaching Circuits | Series RLC

Step 3. Compare *Simulated vs. Real* Measurement Data



Compare Oscilloscope data

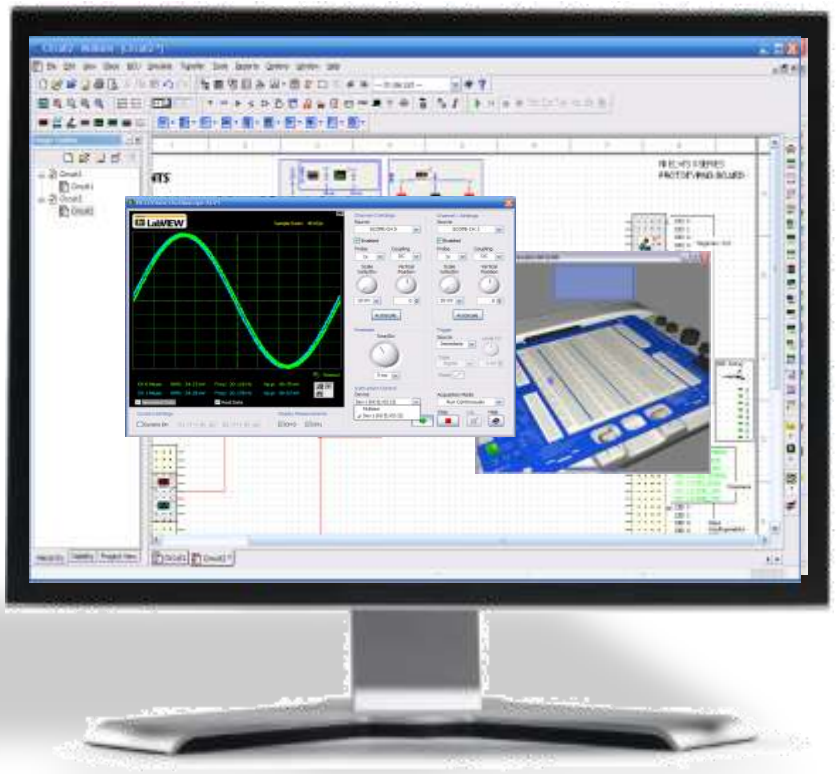


Compare Bodeplotter data

B. Schematic Capture

"Make everything as simple as possible, but not simpler."

– Albert Einstein

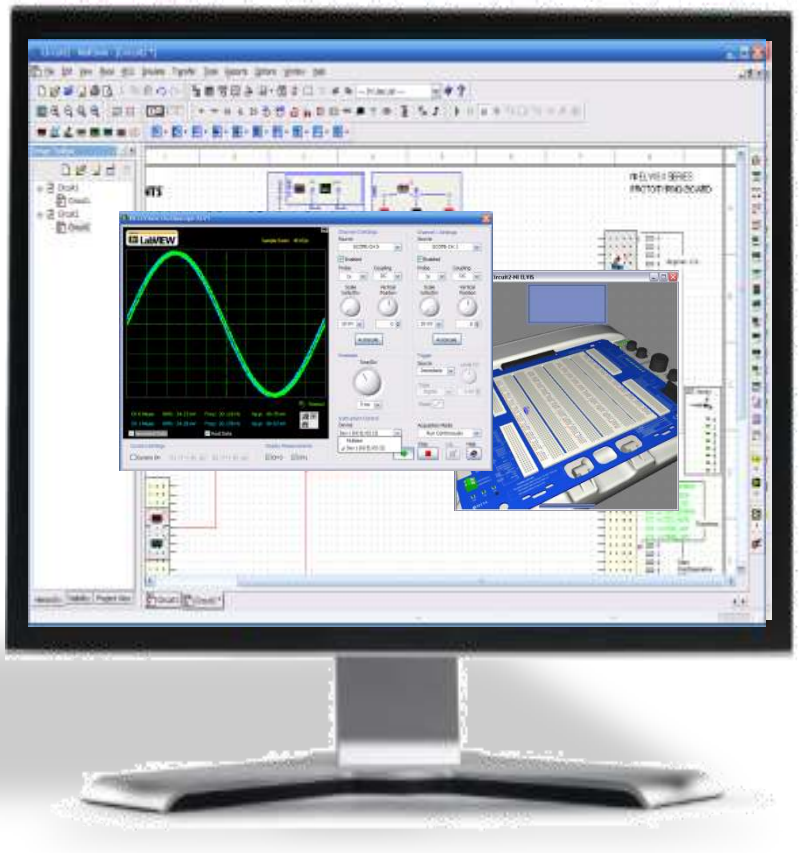


B. What is NI Multisim?

Capture

Simulate

Analysis



Unified environment for teaching circuit theory and design through capture and simulation

B. Schematic Capture with NI Multisim

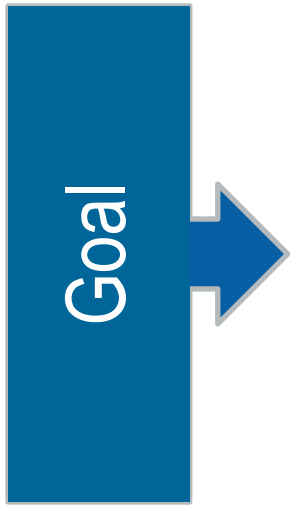
Three steps to schematic capture with Multisim:

- Selecting parts from the database
- Placing parts (locations and orientation)
- Wiring components together



DEMO

Multisim Environment



Objective:

Familiarize with Multisim environment
Select Components from Component
Place components
Wire circuit

Exercise 1

Schematic Capture



Approximate time to complete: 15 minutes

Objective:
Finding and placing parts for 741 Circuit

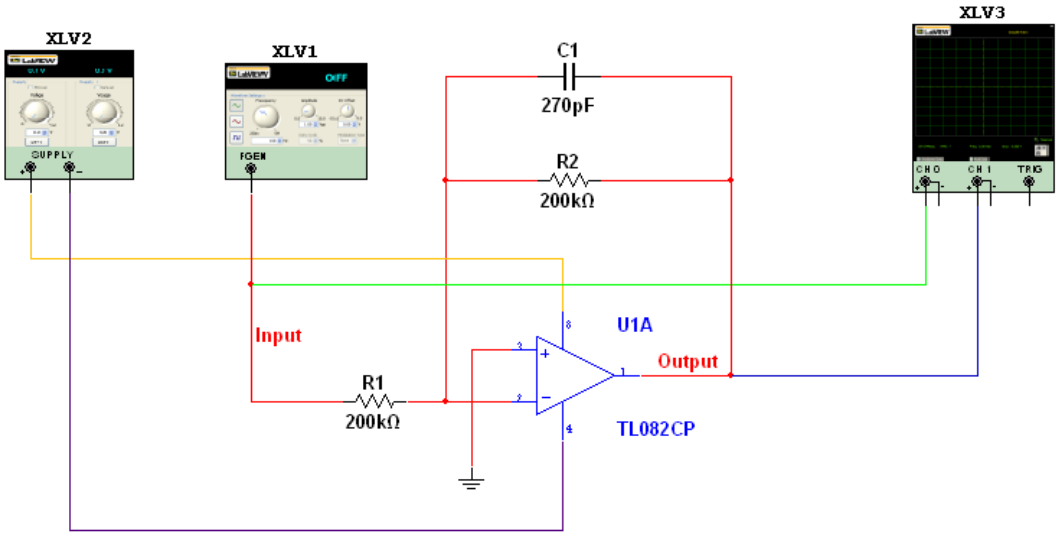
Familiarize with Multisim environment
Select Components
Place components
Wire circuit

C. Simulating Circuits

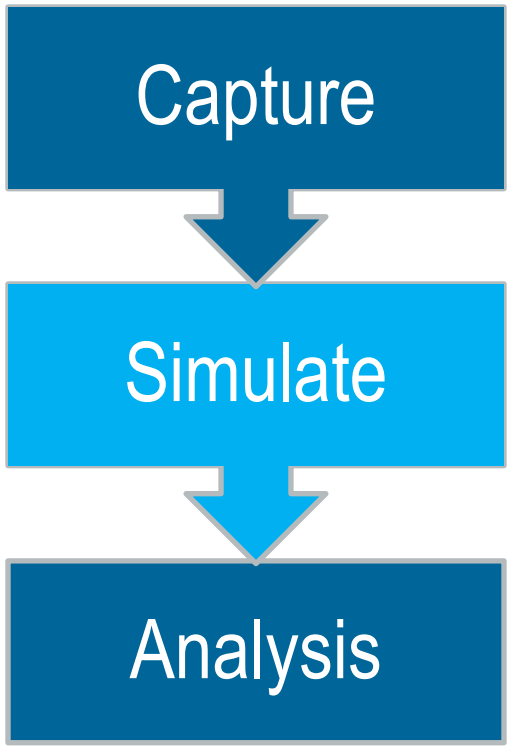
"Computer programs provide a cost-effective means of confirming intended operation"

-Dr. Gordon Roberts

Author of SPICE – Second Edition



C. Simulation



Definition:

Simulation is a mathematical way of emulating the behavior of a circuit.

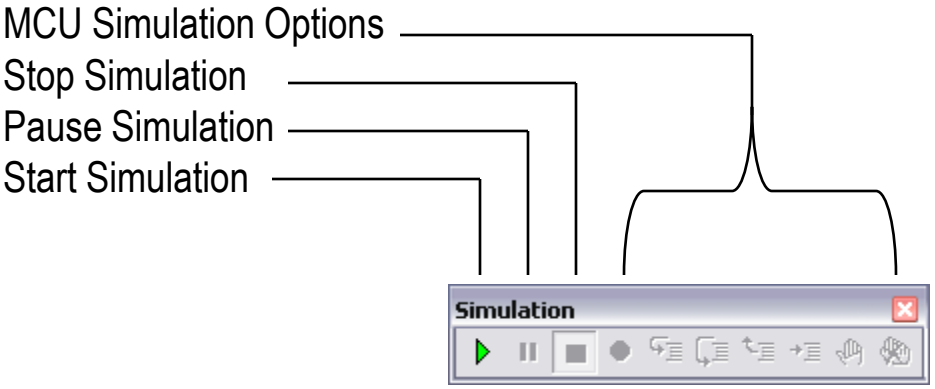
Benefit:

Simulation can determine a circuit's performance without physically constructing the circuit or using actual test instruments.

- incorporates SPICE3F5 and XSPICE at the core of its simulation engine
- non-SPICE-standard Cadence® PSpice® compatibility features

C. Simulation in Multisim

Method 1 – Interactive SPICE Simulation

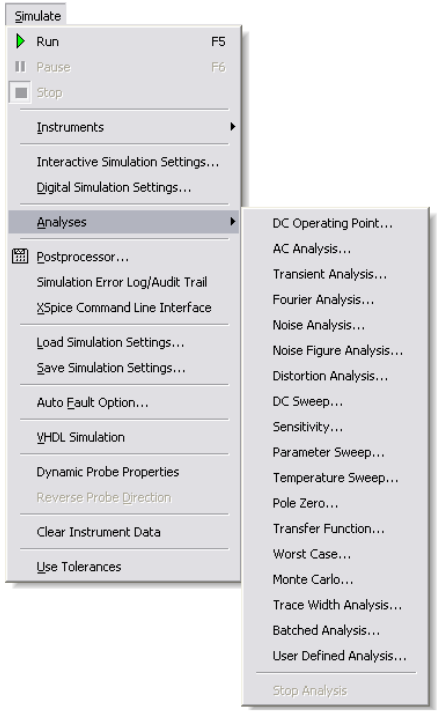


Simulation Toolbar



Interactive Simulation Status

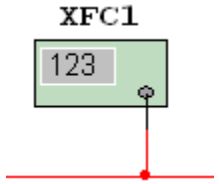
Method 2 – SPICE Analyses



C. Simulation Method 1: Interactive Simulation

- Over 24 instruments are the fastest and easiest way to measure results
- Connect to your circuits just like real equipment
- Double-click the schematic symbol to open the front panel

Frequency Counter



Schematic Symbol

Instruments Toolbar



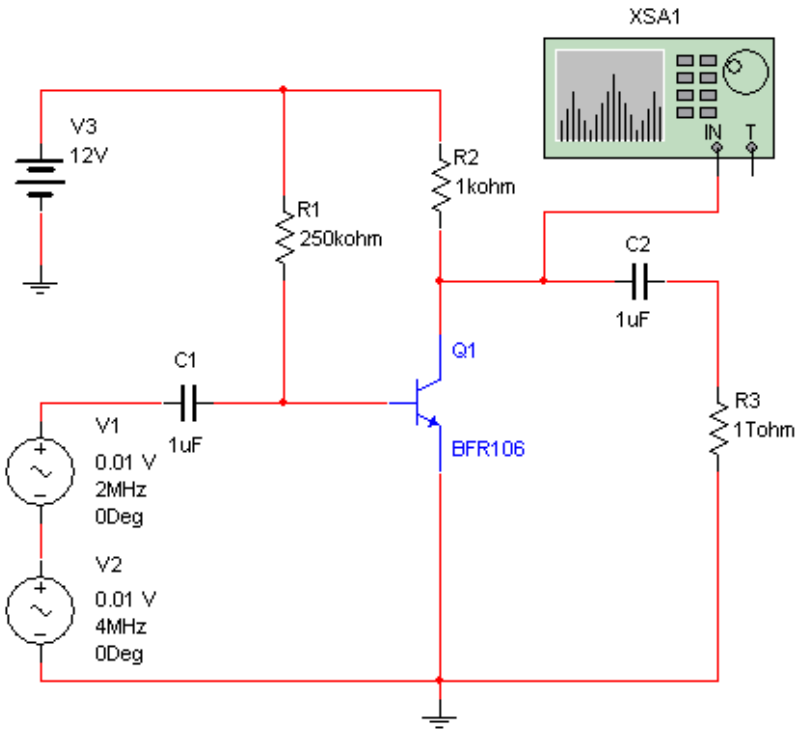
Multimeter
Function Generator
Oscilloscope
Bode Analyzer
Logic Analyzer
Current Probe
LabVIEW Based Instruments
Measurement Probe



Front Panel

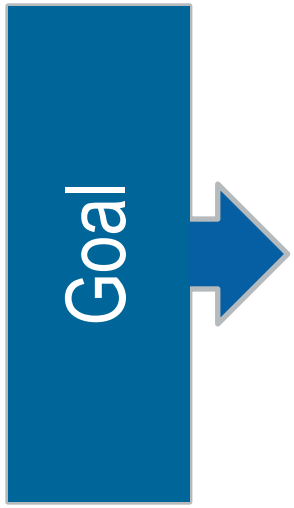
C. Method 1 – Using Instruments in Circuits

1. Access instruments from Design Bar
2. Placing instruments in circuit
   - Multiple Instruments
   - Connect



DEMO

Multisim Simulation Driven Instrumentation



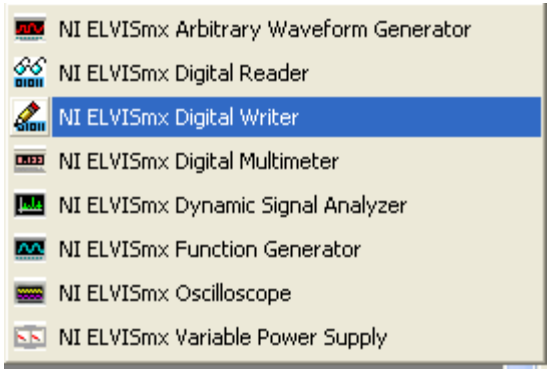
Objective:

Use Multisim instruments to investigate the behavior of the circuit.

Using Multisim instruments and running simulation.

Additional Exercise: NI ELVIS Instruments







Goal

Approximate time to complete: 15 minutes

Objective:
Explore the NI ELVIS II Instruments

3-Wire Current-Voltage Analyzer
2-Wire Current-Voltage Analyzer
Impedance Analyzer
Bode Analyzer
AM Modulation – function generator, oscilloscope arbitrary waveform generator, DSA

Exercise 2

Simulating Circuits

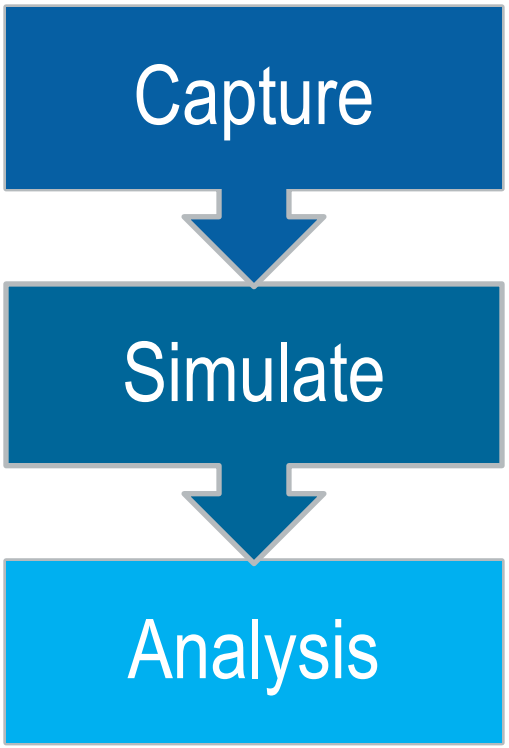


Approximate time to complete: 15 minutes

Objective:
Use Multisim instruments to investigate the behavior of the circuit.

Using Multisim instruments and running simulation.

C. Method 2 – Circuit Analysis

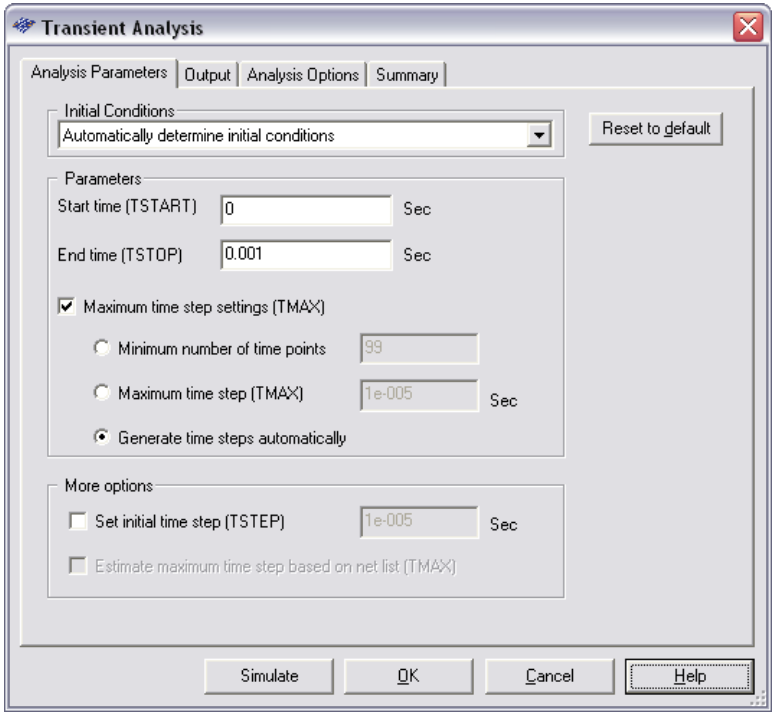


Over 24 SPICE analyses provide advanced simulation capabilities

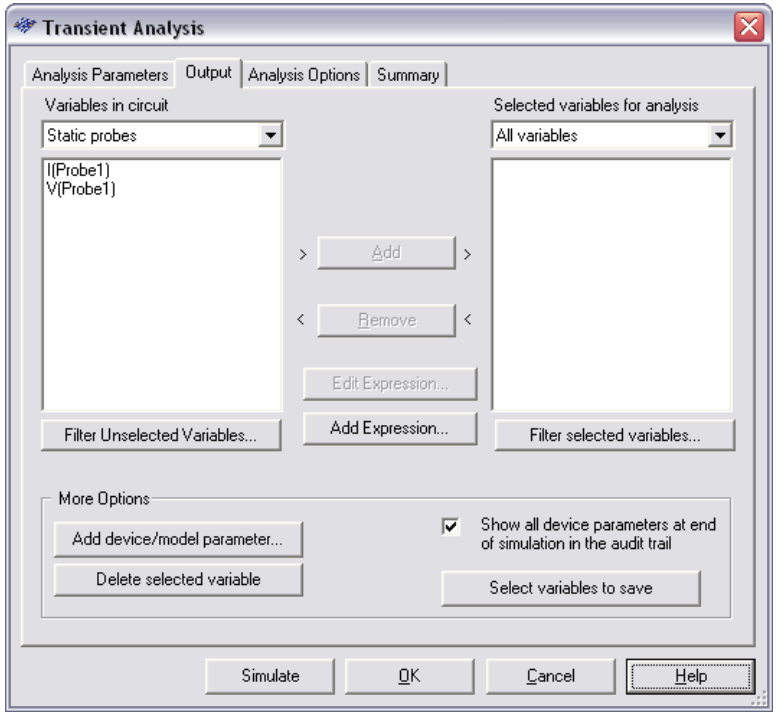
- DC Operating Point
- Transient
- AC Frequency Sweep
- 3 dB Point
- User Defined Analysis
- Fourier
- Noise
- Distortion
- Temperature Sweep
- Model Parameter Sweep
- I-V Analysis
- AC Sensitivity
- DC Sensitivity
- Pole-Zero
- Transfer Function
- Worst Case
- Monte Carlo
- Batched Analysis

C. Method 2 – Circuit Analyses

- Each Analyses has an intuitive configuration wizard



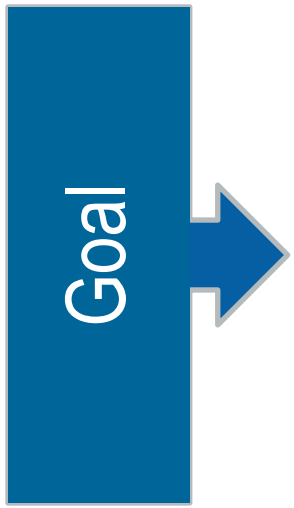
Configuration Dialog



Output Control

DEMO

SPICE Analysis



Objective:

Use Multisim analyses to investigate the behavior of the circuit.

D. Prototyping

"Prototyping produces practical experience from which to judge manufacturability of a design"

-Dr. Gordon Roberts

Author of SPICE – Second Edition

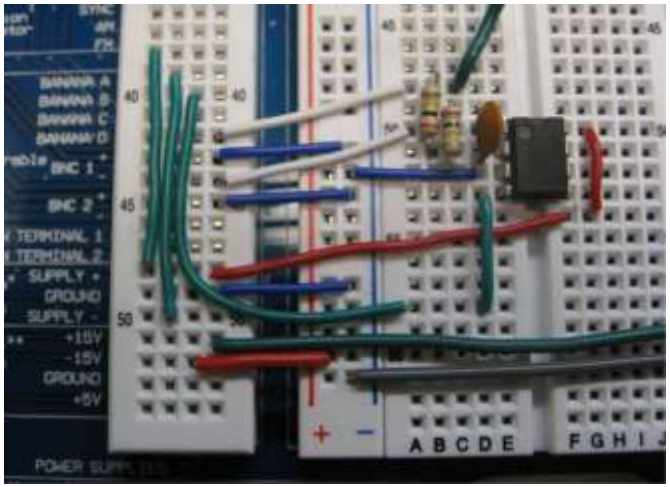


D. Prototyping

- Simulation should never be used exclusively in a design
- Prototyping allows real-world effects to be measured
- Typically breadboards are used in the classroom for prototyping

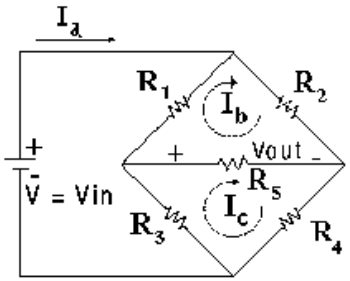


3D Virtual Prototyping



NI ELVIS Prototyping

D. Virtual 3D NI ELVIS







- 3D Breadboard provides risk-free prototyping
- Feedback informs you of successful wiring
- See correlation between schematic and breadboard

D. Prototyping with NI ELVIS

- 12 computer-based LabVIEW Virtual Instruments
- Hi-speed USB plug-and-play connectivity
- Prototyping Breadboard

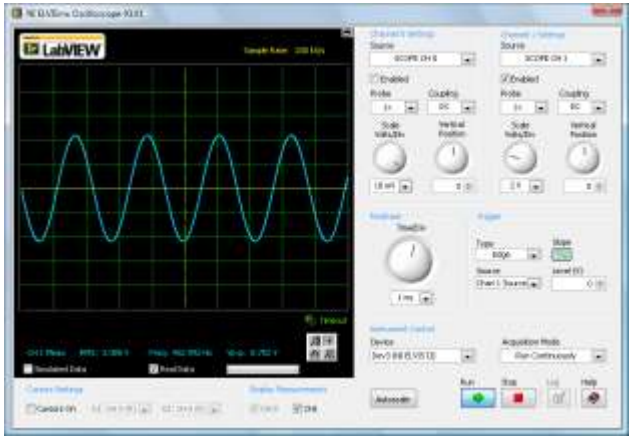


Function Generator





NI ELVIS Workstation

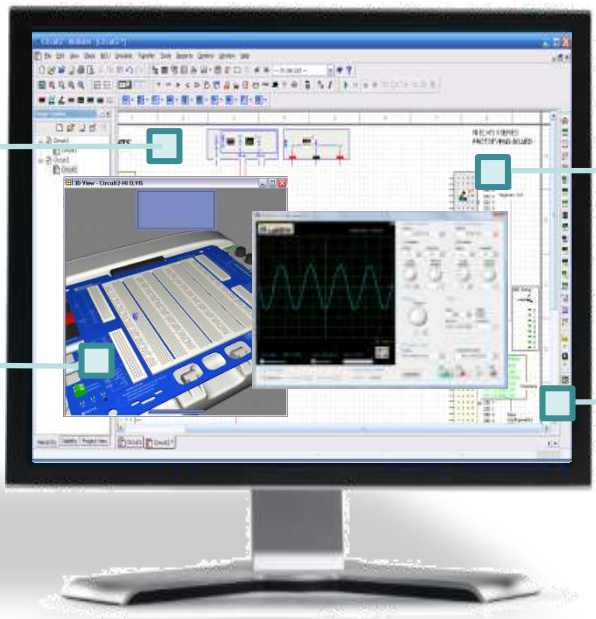


Oscilloscope

D. NI ELVIS II Instruments Inside Multisim

Complete integration of NI ELVIS II inside the Multisim environment facilitates transition of circuit theory to the real-world

NI ELVIS II Schematic
New schematic with NI ELVIS II protoboard pin outs



NI ELVIS II Schematic
Click NI ELVIS II icons on the schematic to access NI ELVIS Instruments

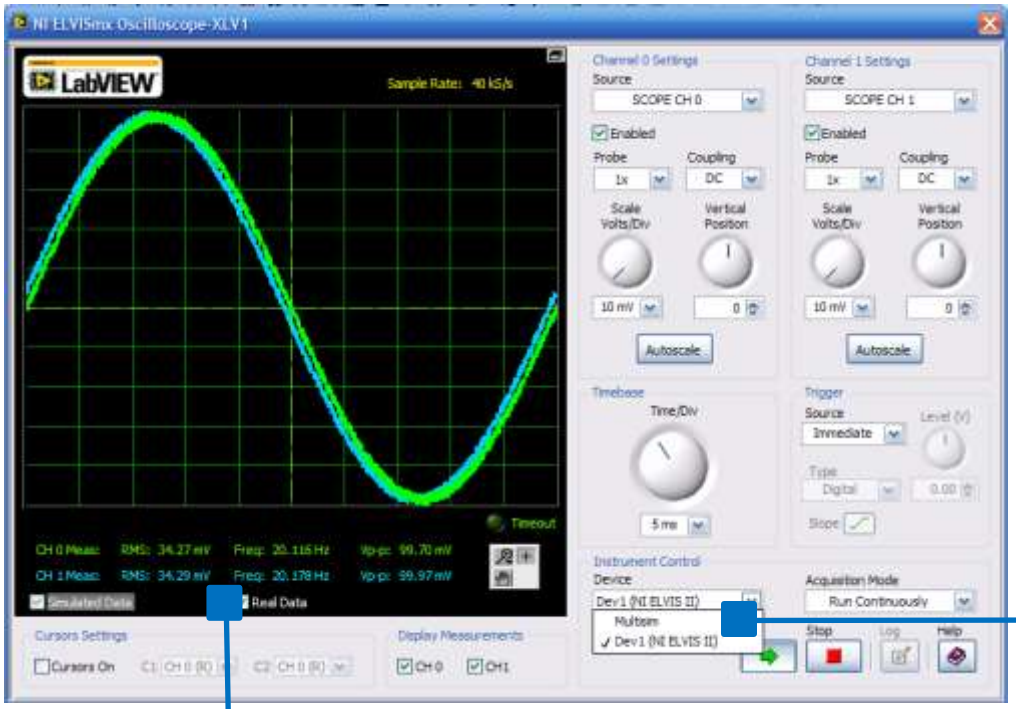


3D NI ELVIS II Updated 3D model and components that allows students to prototype in safe environment

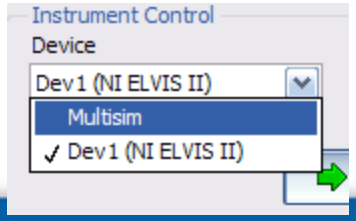
NI ELVIS II Instruments
Access them from the toolbar in addition to other Multisim Instruments

D. Theory and Measurement

Compare simulated data and measured signal on the same instrument



Access NI ELVIS hardware With one click switch between simulated signals and acquiring signals from your NI ELVIS II hardware



NI ELVIS II Instruments Compare simulated Multisim data with measured signals from NI ELVIS II within Multisim

Exercise 3

Prototyping and Measurements



Approximate time to complete: 15 minutes

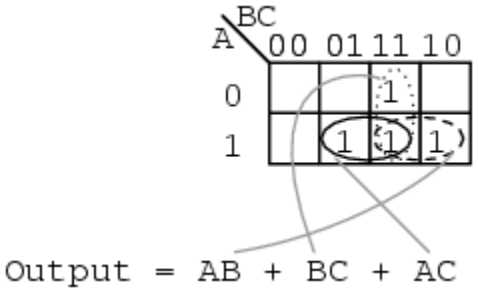
Objective:

DEMO: Explore the 3D Multisim environment.

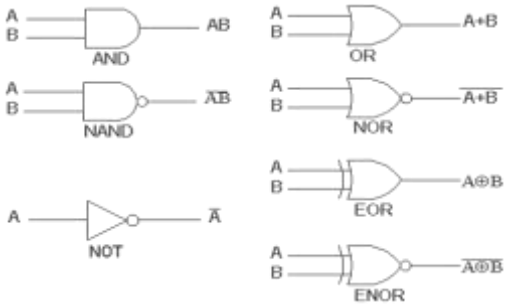
Use NI ELVIS hardware to test and measure design.

Additional Supplement
Digital Electronics & VHDL

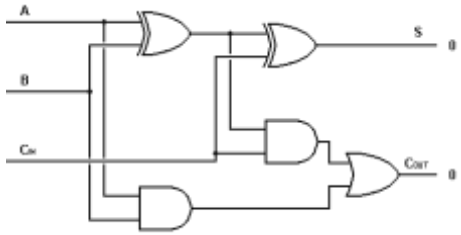
Teaching Digital Electronics | Challenge



Boolean Algebra



Logic Gates



Digital Logic Design

VHDL

```
-- Use: This file defines the top-level of the dest
-- Use With File:

library ieee;
use ieee.std_logic_1164.ALL;
use ieee.numeric_std.ALL;

library work;
use work.test_vhdl_pkg.ALL;

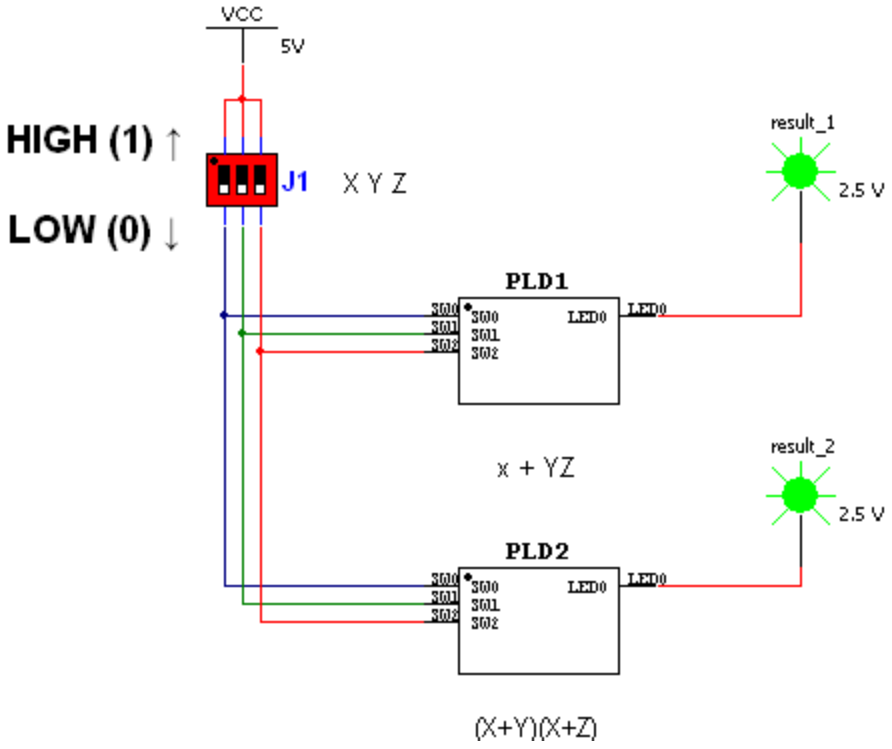

entity binary_adder is
        port (

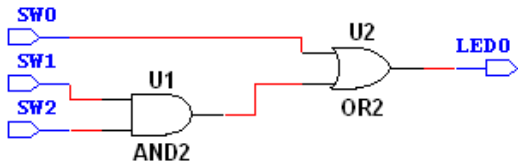
                LED0 : out std_logic;
                LED1 : out std_logic;
                LED2 : out std_logic;
                SW0 : in std_logic;
                SW1 : in std_logic;
                SW2 : in std_logic;
                SW3 : in std_logic;
                sys_clk_pin : in std_logic
        );
end binary_adder;
```

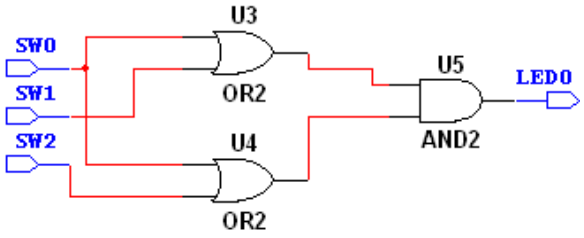
Teaching Digital | Distributive Property of AND

Step 1: Capture, Simulate and understand a design in NI Multisim

$$X + YZ = (X + Y)(X + Z)$$

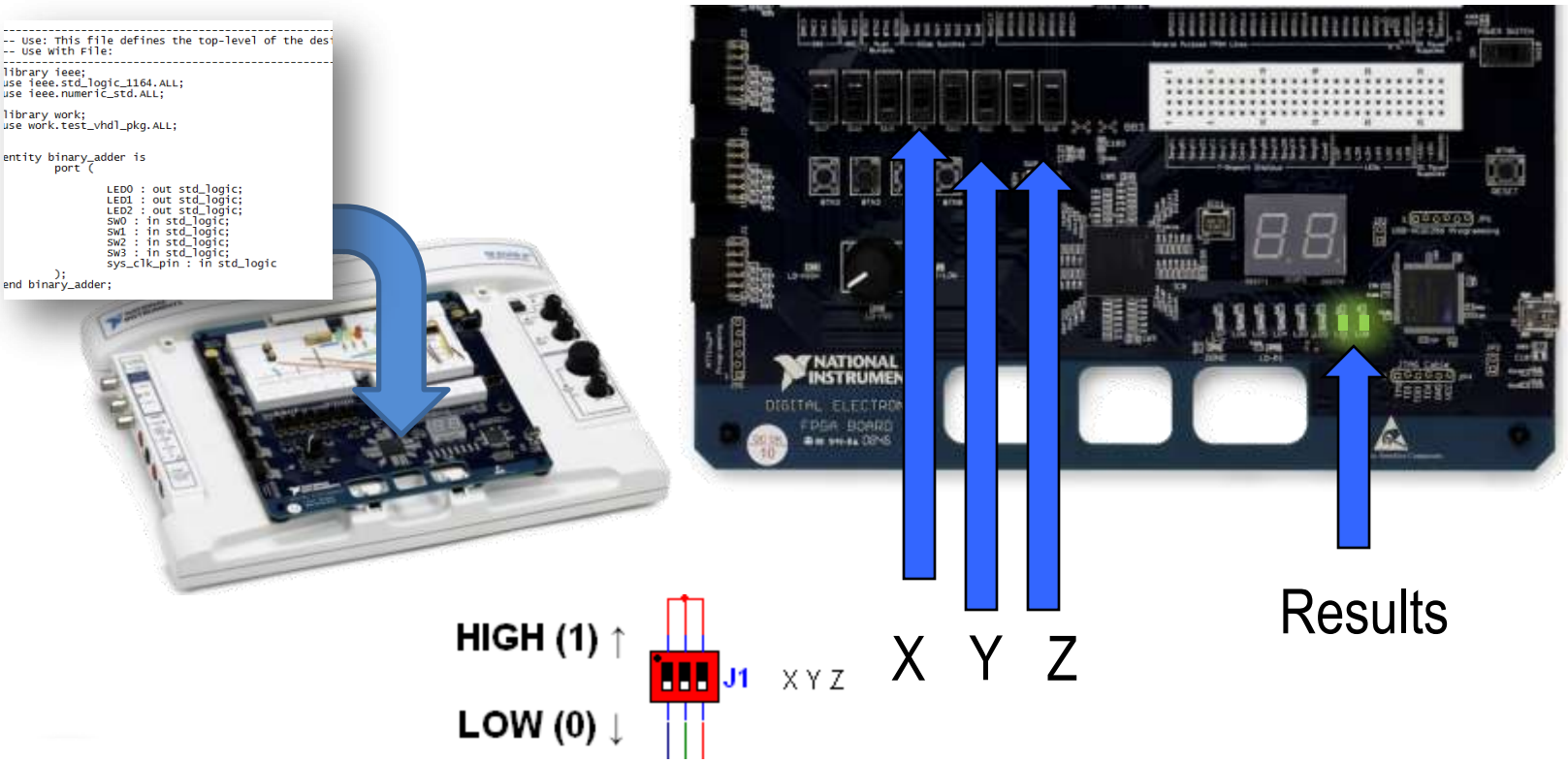






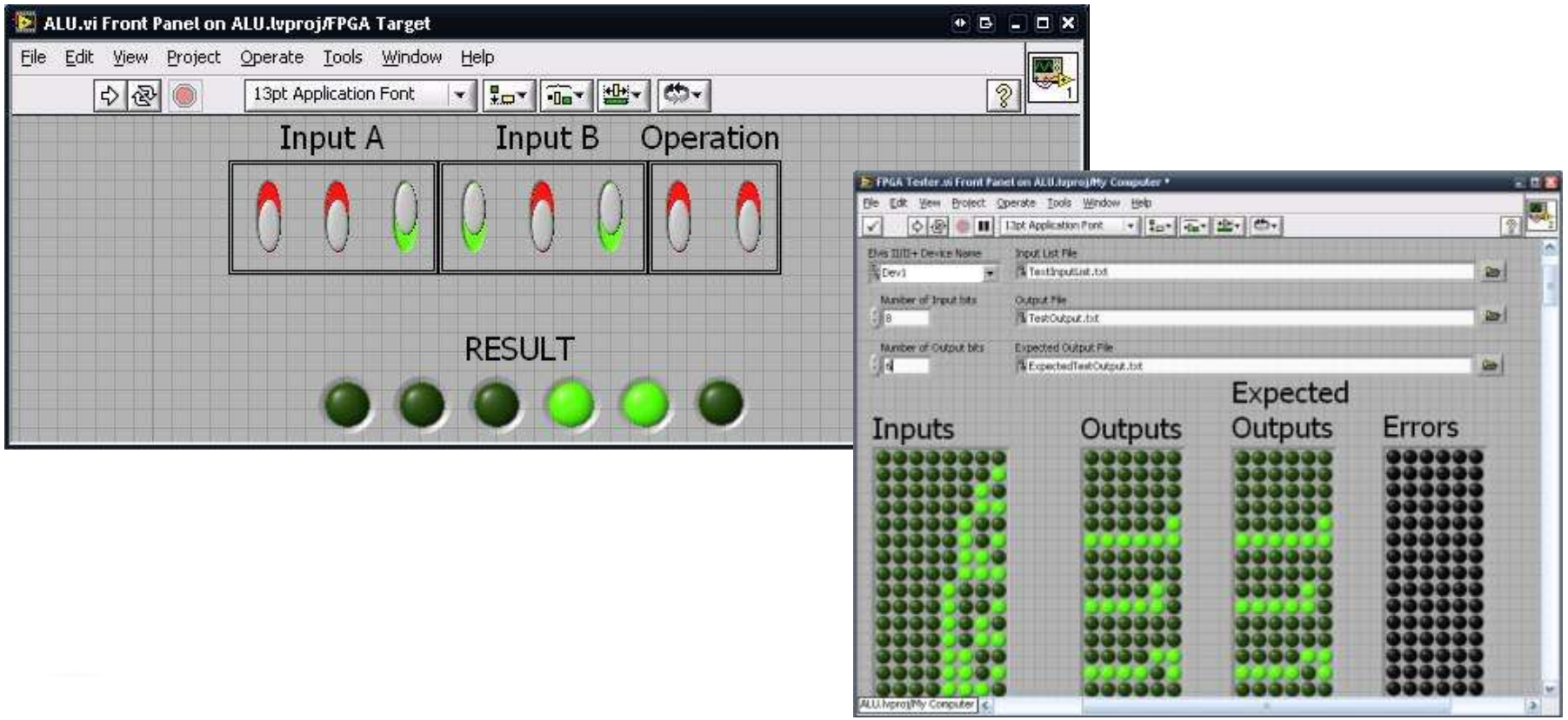
Teaching Digital | Distributive Property of AND

Step 2: Target FPGA with VHDL and measure signals with NI ELVIS

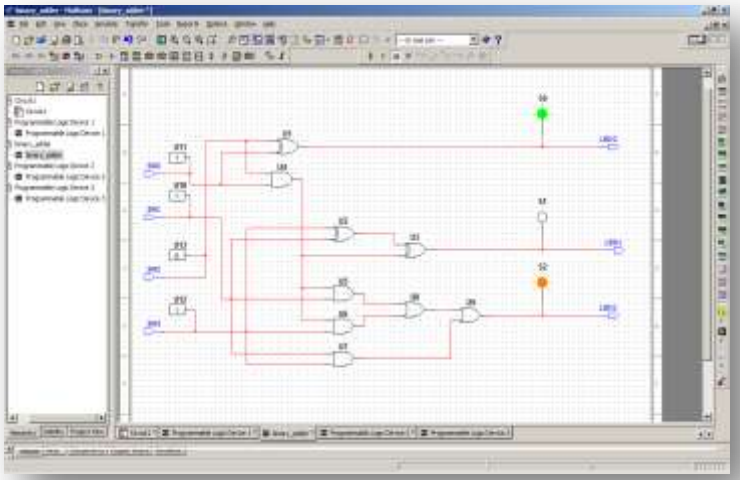


Teaching Digital | Distributive Property of AND

Step 3. Compare *Simulated vs. Real* Measurement Data



Programmable Logic Design in Multisim





```
-- Use: This file defines the top-level of the dest
-- Use With File:
-------------------------------------------------------------
library ieee;
use ieee.std_logic_1164.ALL;
use ieee.numeric_std.ALL;

library work;
use work.test_vhdl_pkg.ALL;


entity binary_adder is
        port (

                LED0 : out std_logic;
                LED1 : out std_logic;
                LED2 : out std_logic;
                SW0 : in std_logic;
                SW1 : in std_logic;
                SW2 : in std_logic;
                SW3 : in std_logic;
                sys_clk_pin : in std_logic
        );
end binary_adder;
```

Programmable Logic Design from Schematic

- Generate raw VHDL from a schematic drawn in Multisim
- 100+ basic components
- VHDL code can be use to target programmable logic devices

NI Digital Electronics FPGA Board

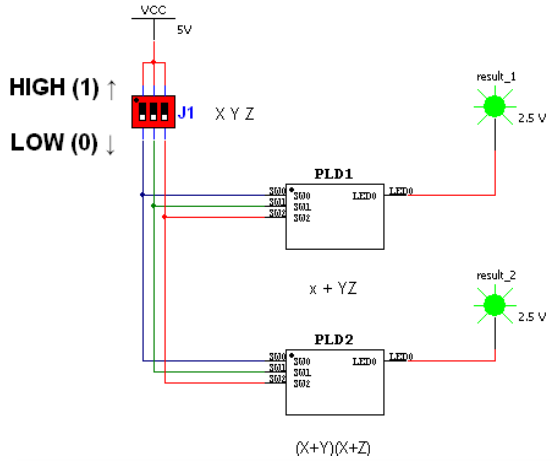
Digital electronics and design board with field-programmable gate array (FPGA)



- Hands-on learning thorough implementation
- Student interaction with onboard displays, switches and buttons
- Programmable with LabVIEW and Xilinx ISE tools

Teaching Digital Concepts

1. Capture a digital design
2. Use interactive simulation to reinforce concepts
3. Use simulation driven instruments
4. Export design to VHDL





Exercise

Programmable Logic Schematic



Approximate time to complete: 20 minutes

Objective:

Use the PLD Schematic to explore the digital circuit.

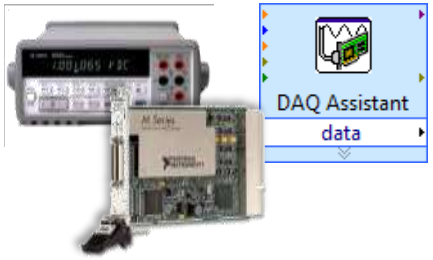
E. Measurement & Analysis

"In theory, theory and practice are the same. In practice, they are not."

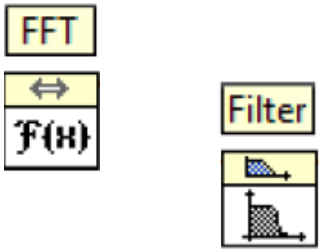
- "Yogi" Berra



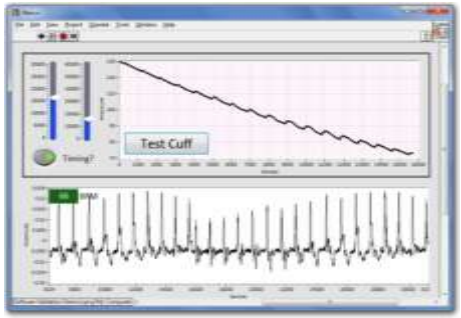
Elements of Engineering



Hardware APIs



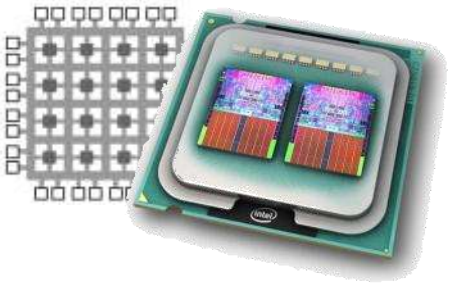
Analysis Libraries



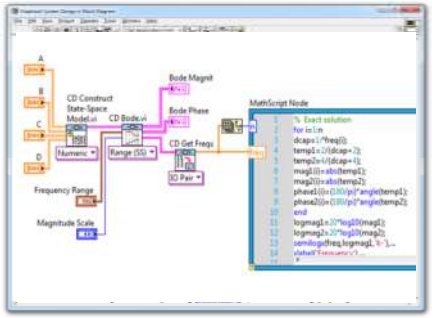
Custom User Interfaces



Deployment Targets

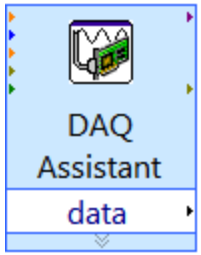


Technology Abstractions

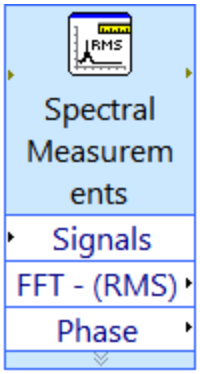


Programming Approaches

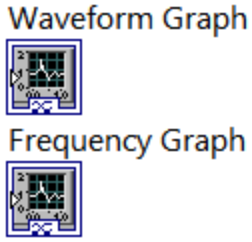
Compatible Elements



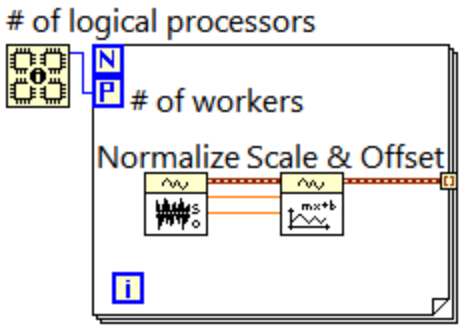
Hardware APIs



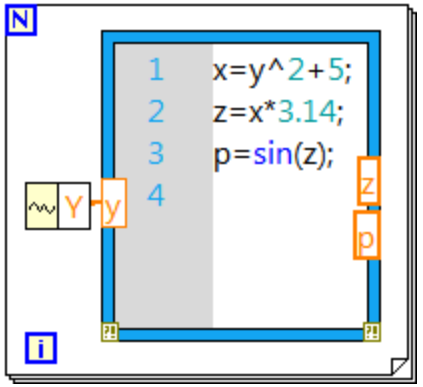
Analysis Libraries



Custom User Interfaces

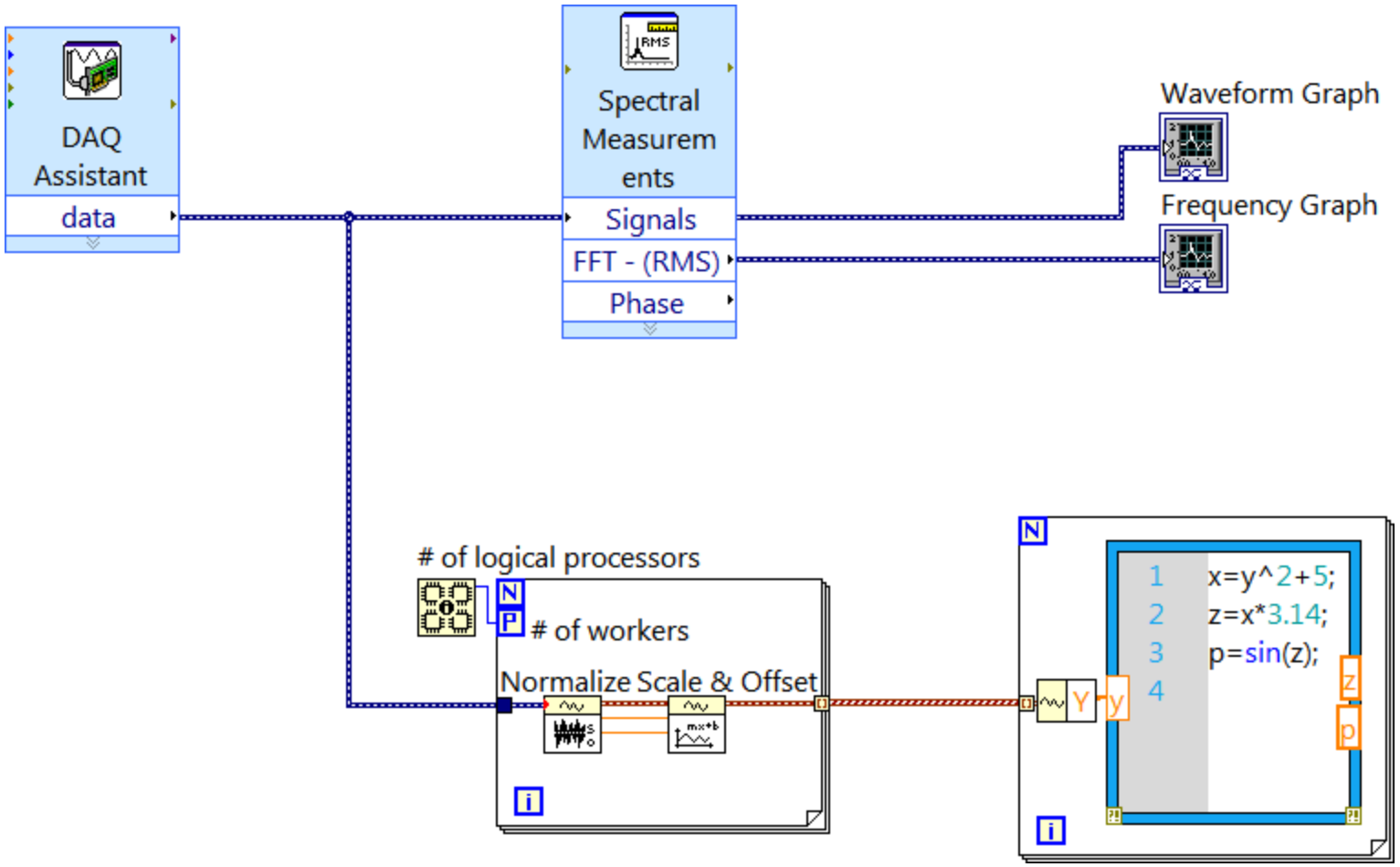


Technology Abstractions



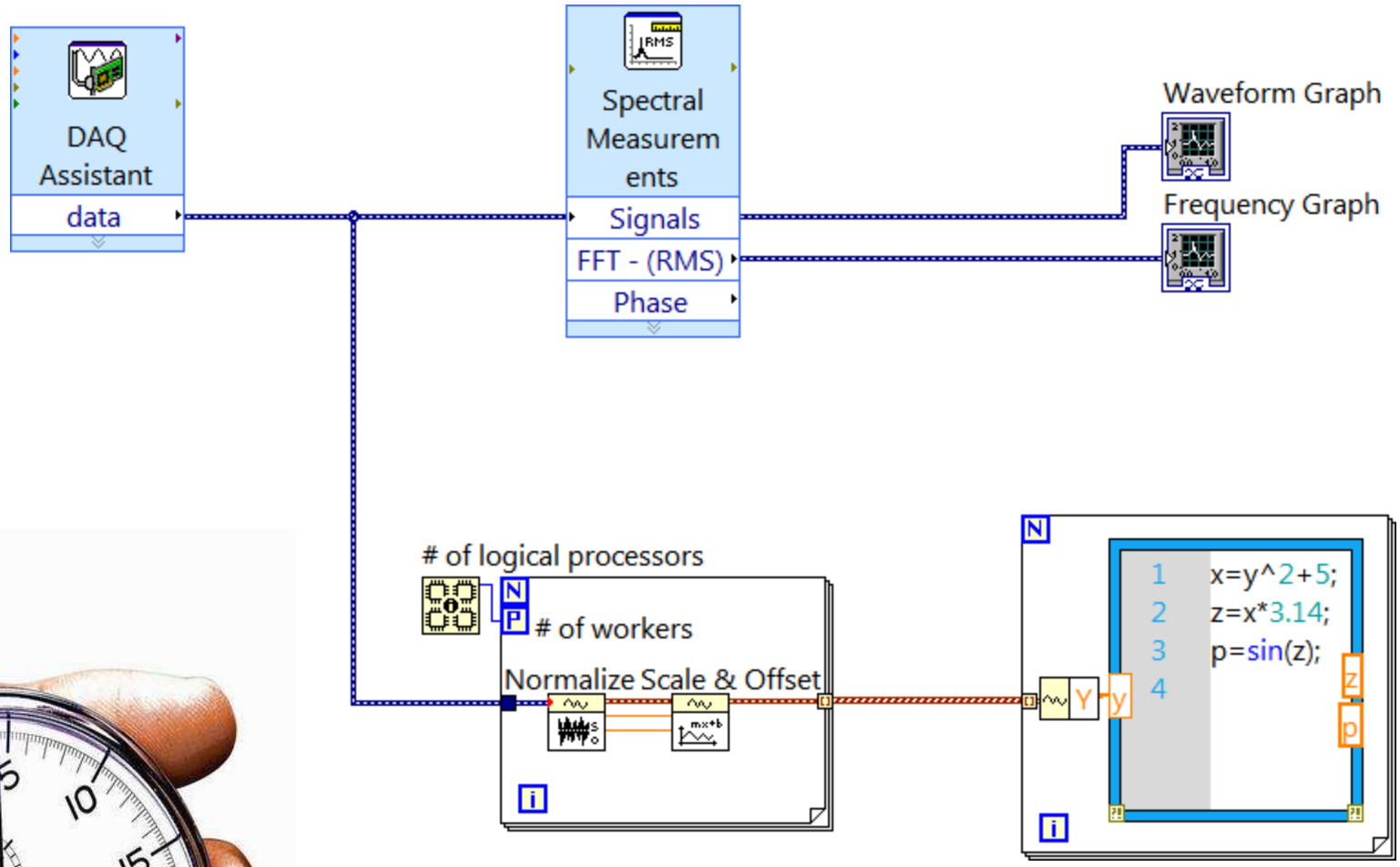
Programming Approaches

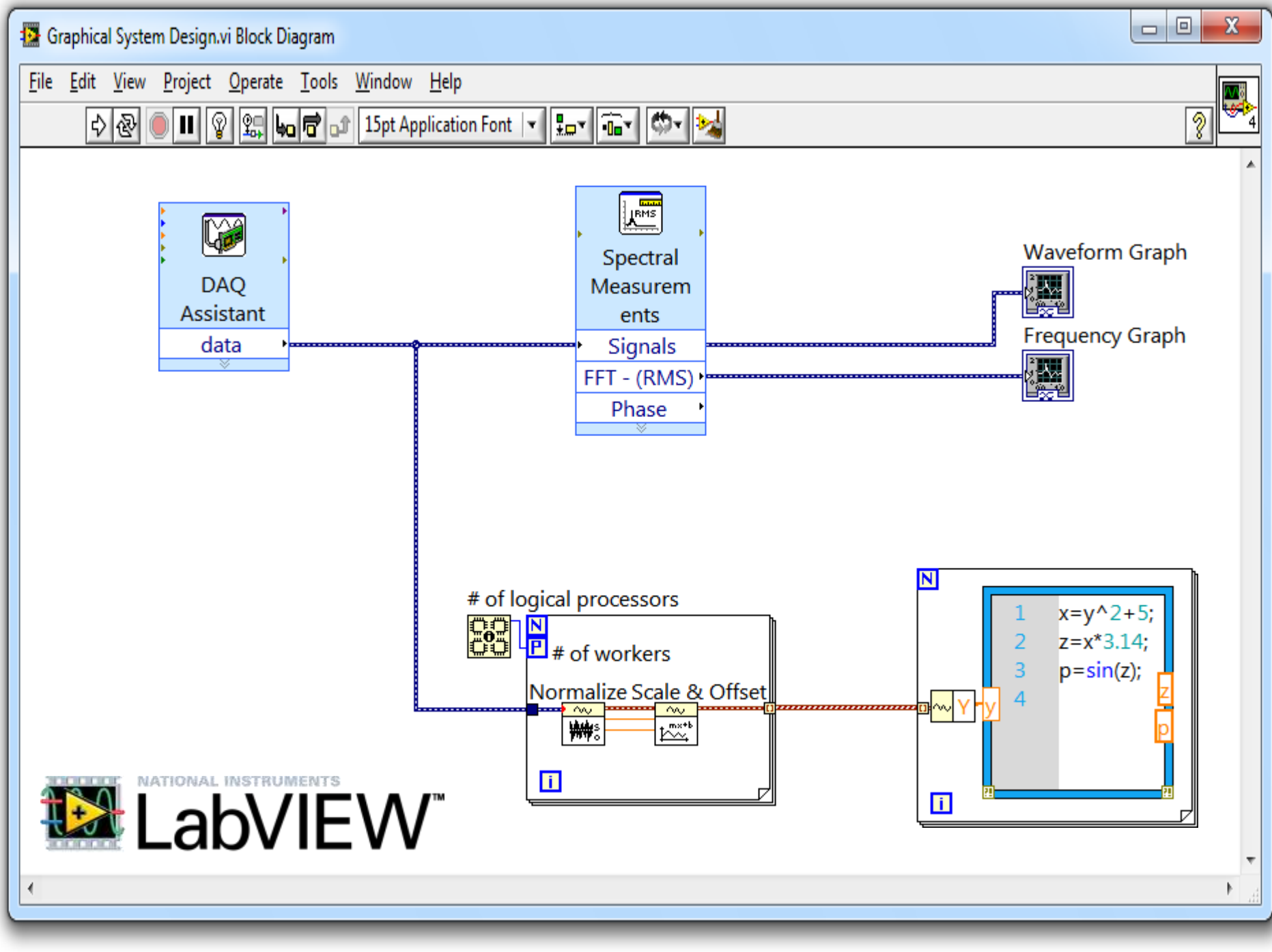
Easily Combined



In the Time Given






Graphical System Design.vi Block Diagram
File Edit View Project Operate Tools Window Help
15pt Application Font
DAQ Assistant
data
Spectral Measurements
Signals
FFT - (RMS)
Phase
Waveform Graph
Frequency Graph
of logical processors
of workers
Normalize Scale & Offset
1 x=y^2+5;
2 z=x*3.14;
3 p=sin(z);
4
NATIONAL INSTRUMENTS
LabVIEW™

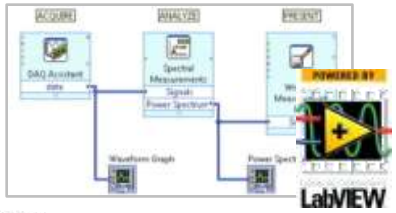
National Instruments Academic Program

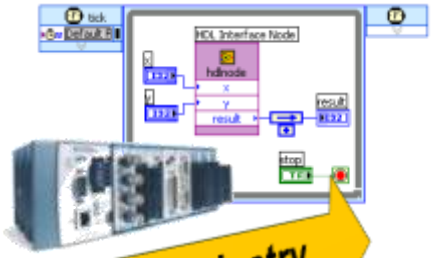
From Kindergarten to Rocket Science



LEGO® MINDSTORMS® NXT
powered by LabVIEW





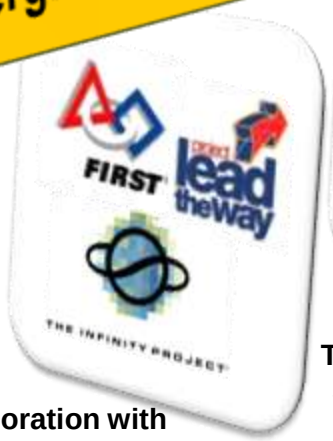


Kindergarten - 12

University

Research

Industry



Collaboration with
Leading STEM Programs



Textbook & Author
Support Program



In More Than 7500
Universities in 110 Countries

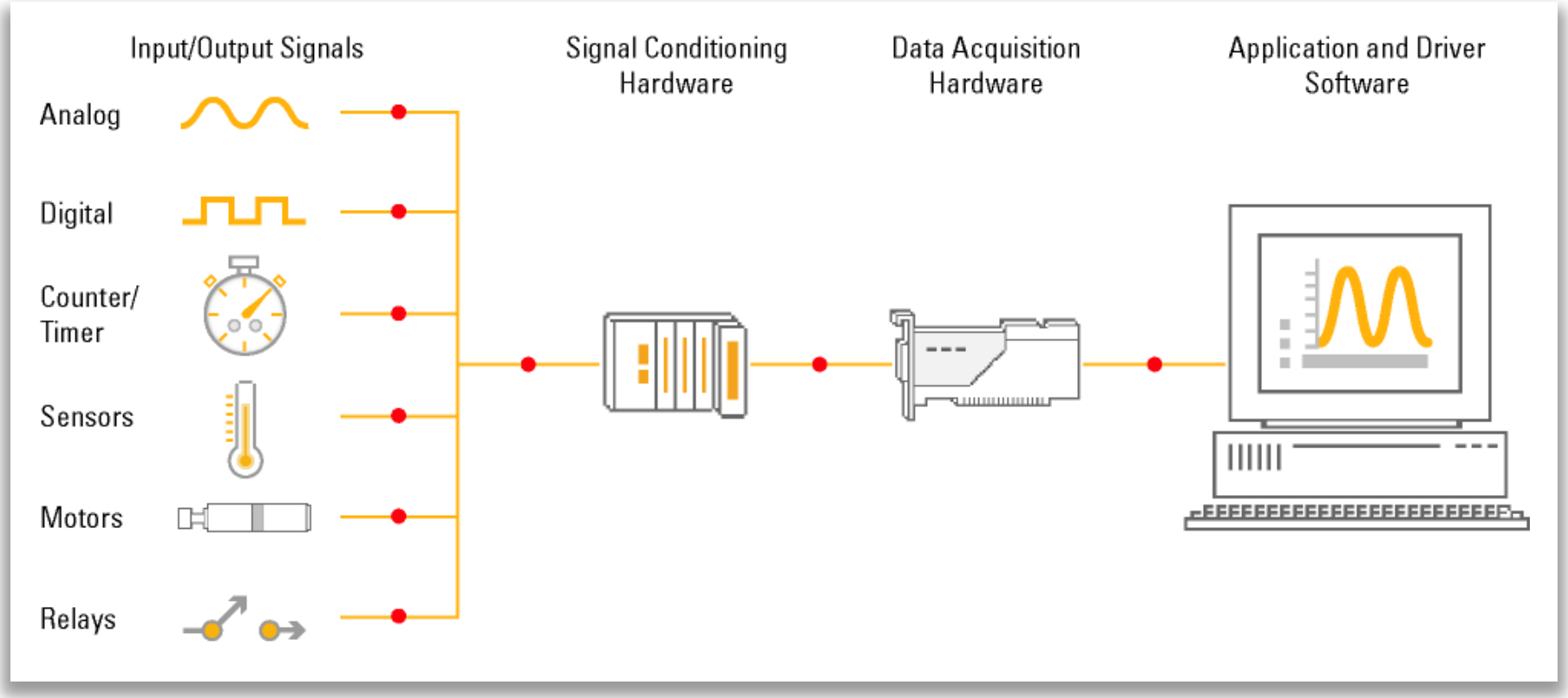


LabVIEW Academy
Certification

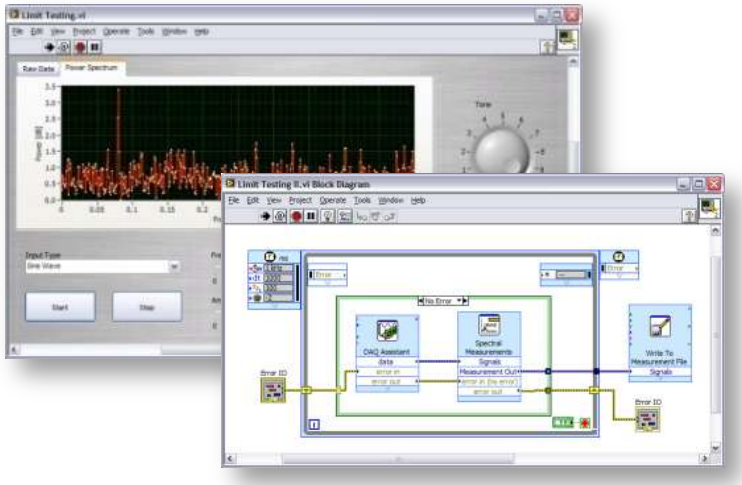


Collaborative
Research

Introduction to Data Acquisition



NI DAQ Platform





USB



Wireless



PCI



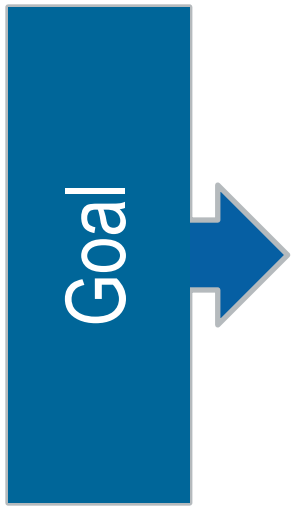
CompactDAQ



PXI

DEMO

LabVIEW Environment

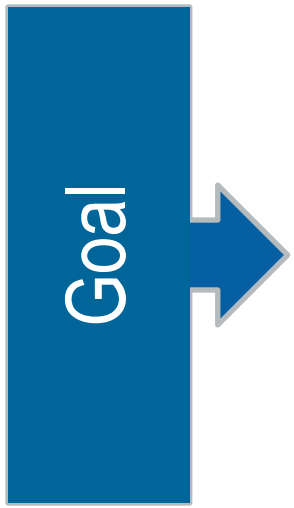


Objective:

Introduce the LabVIEW environment.

Exercise

Data Acquisition with NI ELVIS & LabVIEW



Approximate time to complete: 20 minutes

Objective:

Acquire signals from your FGEN with Analog Input Channels using LabVIEW

- Express Vis
- Low :Level DAQmx API

Exercise

Concept of While Loop



Approximate time to complete: 20 minutes

Objective:

Explore the concept of a while loop in LabVIEW using NI ELVIS as the hardware platform that will be acquiring signals.

Exercise (Optional)

Audio Equalizer

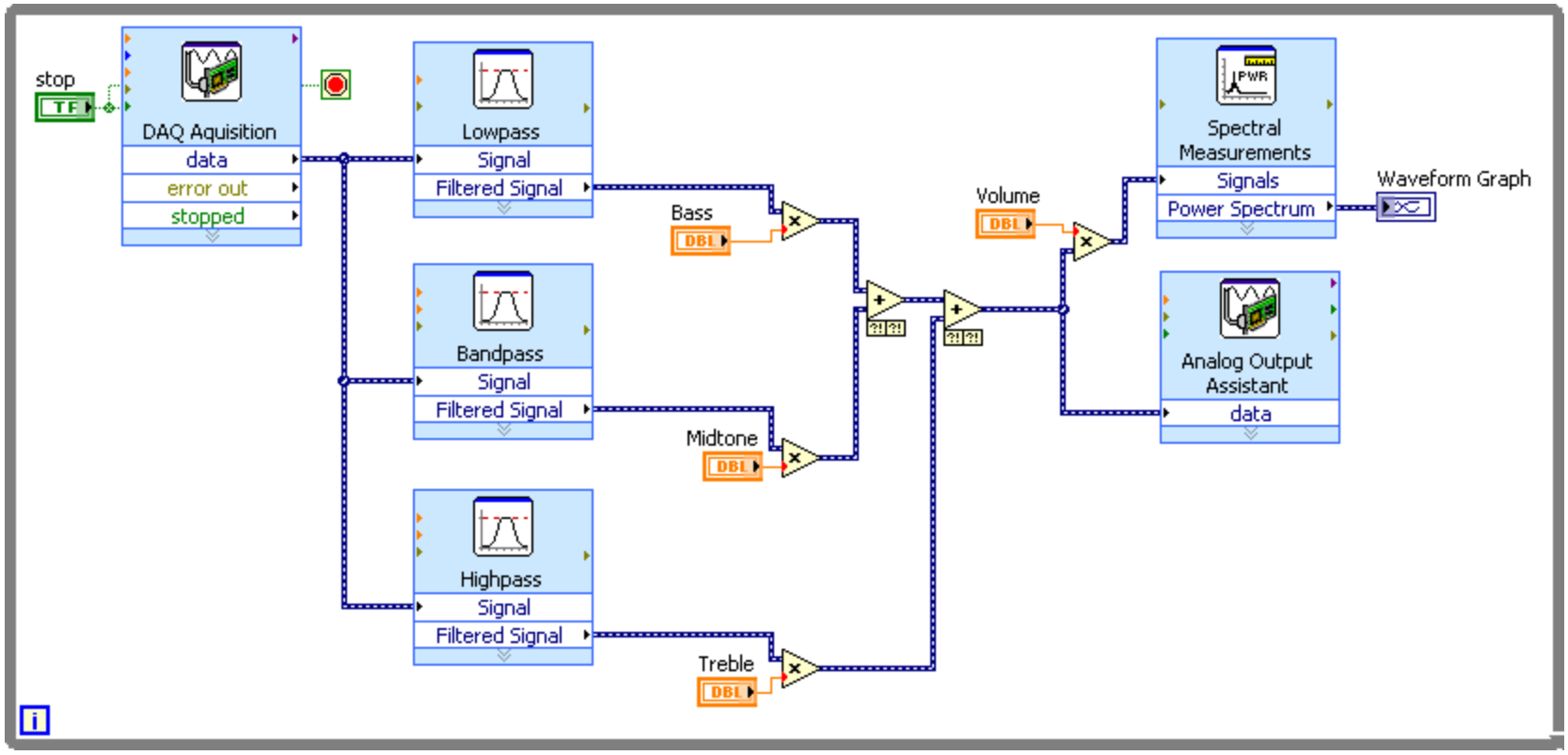


Approximate time to complete: 30 minutes

Objective:

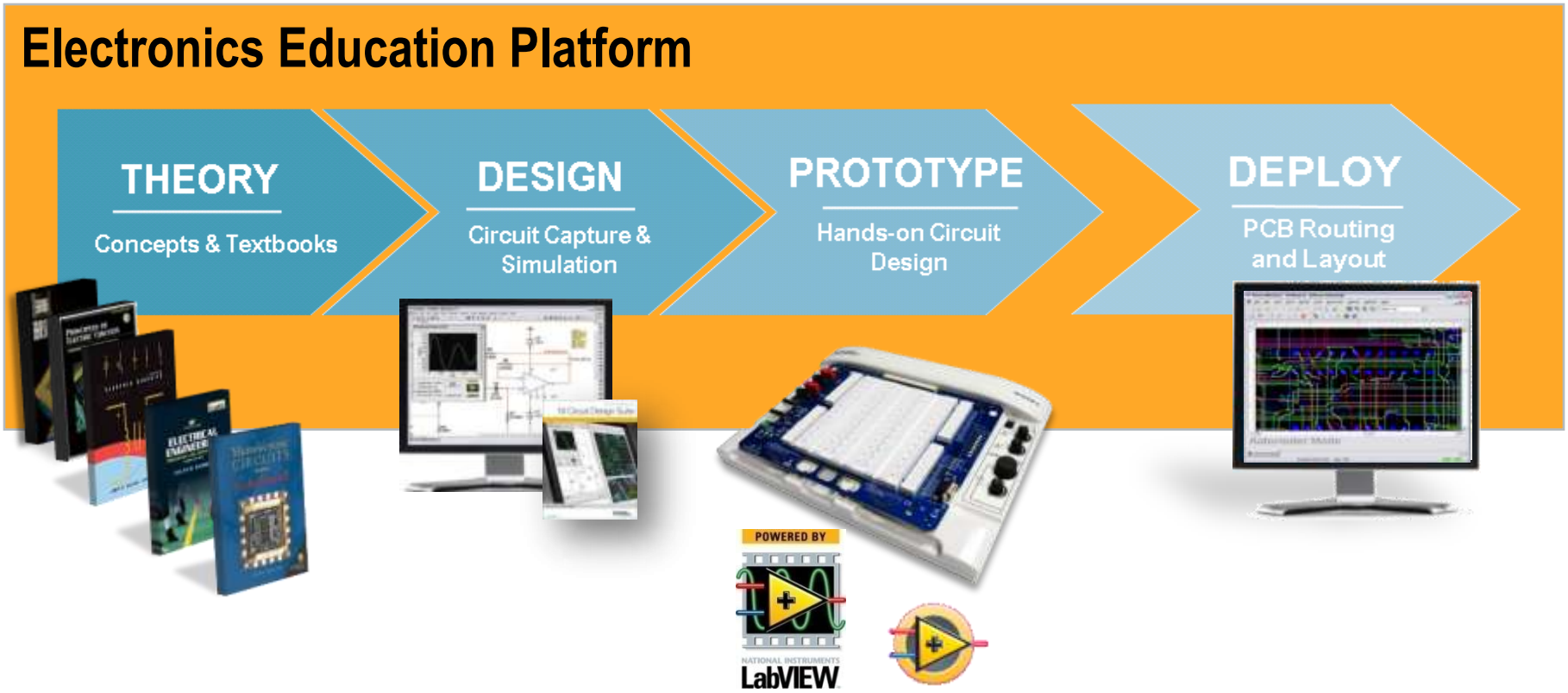
Explore the concepts of data acquisition, signal processing, signal analysis, data generation and data visualization through an Audio Equalizer application.

Audio Equalizer



Conclusion

National Instruments tools address challenges of circuit education



For more information



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- **Curricula**: Practical Teaching Ideas, Teaching RF in Multisim, NI ELVIS II
- **Tutorials and whitepapers**: How to tutorials and simulation fundamentals